

#### **TUTORIAL 4: DNA Translation**

Q1. Define translation in the context of molecular biology. (2 marks)

- Translation is the process by which the sequence of a messenger RNA (mRNA) is decoded by a ribosome to synthesize a specific polypeptide or protein

Q2. What are the start and stop codons in translation? (2 marks)

- start codon which is usually AUG, is the first codon of an mRNA that tells ribosome where to start making the protein
- stop codons are UAA, UAG, and UGA which signal the termination of translation

Q3. Name the three sites on the ribosome involved in translation. (1 mark)

- A-site (Aminoacyl site)
- P-site (Peptidyl site)
- E-site (Exit site)

Q4. Explain the significance of the Shine-Dalgarno sequence in bacterial translation. (3 marks)

- Shine-Dalgarno sequence matches with a part of the ribosome to the mRNA
- make sure the start codon (AUG) is in the right spot to begin translation
- ensures the ribosome starts making the protein correctly and efficiently

Q5. Describe the role of tRNA in protein synthesis. (3 marks)

- carrying amino acids to the ribosome
- matching codons on mRNA with its anticodon to ensure the right amino acids is added
- helping the ribosome link amino acids to form a protein chain

Q6. Given the mRNA sequence 5'-AUG GUC GCU AAA AGU UGA-3', determine the corresponding amino acid sequence. (3 marks)

- AUG(met) GUC(val) GCU(ala) AAA(lys) AGU(ser) UGA(no amino acid)
- Met-Val-Ala-Lys-Ser

Q7. How does tmRNA rescue stalled ribosomes? (3 marks)

- tmRNA carries alanine, which is part of stop codon
- when it finds a stalled ribosome, it binds near the defective mRNA
- tmRNA provides the necessary stop codon to free the trapped ribosome, allowing translation to terminate properly

Q8. Compare the roles of initiation factors and elongation factors in translation. (4 marks)

| Initiation Factors                                | Elongation Factors                                |
|---------------------------------------------------|---------------------------------------------------|
| Position the first tRNA                           | Assist ribosome movement                          |
| Assemble the initiation complex                   | Facilitate peptide chain growth                   |
| Active during the initiation phase of translation | Active during the elongation phase of translation |

Q9. Analyze the importance of the central dogma in understanding protein synthesis. (4 marks)

- the central dogma defines the flow of genetic information from DNA to RNA and then to protein, explaining how genetic instructions are used to produce proteins
- the central dogma connects an organism's genotype to its phenotype by showing how the genetic code in DNA determines the structure and function of proteins, which influence traits